COVID-19 vaccine efficacy from clinical trial counts

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Scientific question: What is the posterior uncertainty on vaccine efficacy when efficacy is inferred from case counts in randomized vaccine trials?

Data: Use official FDA clinical-review documents for COVID-19 vaccines. Use with the Pfizer–BioNTech/Comirnaty BLA clinical review memorandum, which reports trial counts for vaccine and placebo groups. FDA Comirnaty clinical review memorandum: `https://www.fda.gov/media/152256/download`

Models:

- **Model 1.** independent binomial risks in vaccine and placebo arms,

$$Y_V \sim \text{Binomial}(n_V, p_V), \qquad Y_P \sim \text{Binomial}(n_P, p_P), \tag{1}$$

  where Y_V (Y_P) is the number of cases in the vaccine (placebo) group made up of n_V (n_P) people, p_V (p_P) is the risk of disease in the vaccine (placebo) arms. The vaccine efficacy is defined as

$$\text{VE} = 1 - \frac{p_V}{p_P}. \tag{2}$$

- **Model 2.** logistic model with a treatment indicator,

$$Y_i \sim \text{Binomial}(n_i, p_i), \qquad \text{logit}(p_i) = \alpha + \beta x_i, \tag{3}$$

  where $x_i = 1$ for the vaccine arm and $x_i = 0$ for the placebo arm. In this model

$$p_P = \text{logit}^{-1}(\alpha), \qquad p_V = \text{logit}^{-1}(\alpha + \beta), \tag{4}$$

  and therefore

$$\text{VE} = 1 - \frac{\text{logit}^{-1}(\alpha + \beta)}{\text{logit}^{-1}(\alpha)}. \tag{5}$$

Required output:

- Extract the numbers of participants and COVID-19 cases in vaccine and placebo arms from the FDA document.
- Infer the posterior distribution of VE and report a 95% credible interval.
- Compare the binomial-risk model and the logistic-treatment model.
- Compute the posterior predictive distribution of cases in a hypothetical trial with 10^4 participants per arm.

Introduction and Data

COVID-19 is the infectious disease caused by SARS-CoV-2. It can cause severe respiratory symptoms, pneumonia, respiratory failure, organ failure, and even death. In this context, the vaccine BNT162b2 is indicated for preventing the disease in individuals of 16 years of age and older, with a vaccine regime that consists of two doses [1].

Vaccine efficacy (VE) is measured as one minus the incidence rate ratio (IRR) in the vaccinated group compared with the unvaccinated group [1, 2]. There are variety of methods to estimate confidence intervals for the IRR from the frequentist point of view. However, these struggle when the number of infections in the vaccinated group is low. In this case the Bayesian approach is better for estimating VE and one of its most common measure is based on the comparison of two binomial proportions [2]. This is in fact the case of both implemented models in the present work.

The main study for the efficacy of this vaccine was a randomized trial consistent of 44165 participants (22085 receiving BNT162b2 and 22080 receiving placebo). As expected, efficacy is measured by observing the positive cases of COVID-19, which of course requires the participant's follow-up in the study. It is important to highlight that not all randomized participants are included in the efficacy analyses. In fact, participants can be excluded for not meeting all eligibility criteria despite randomization, for not receiving a dose or not receiving it in the predefined window, also for voluntary withdrawal, or many other different reasons [1].

For evaluating vaccine efficacy the FDA memorandum reports separately depending on the prior SARS-CoV-2 infection status (see Table 14 and 15 of [1]). We chose to work with the data of participants both with or without evidence of prior infection. According to this:

- The total number of participants is 42257 (21047 receiving BNT162b2 and 21210 receiving placebo).
- The participants with confirmed COVID-19 occurring at least 7 days after dose 2 are 935 (81 cases in the BNT162b2 group versus 854 cases in the placebo one).
- The number of participants who were indeed monitored during all the surveillance time is not the same as the total number of participants (20533 in the vaccine and 20595 in the placebo).

Participants	**Vaccine**	**Placebo**
N1	81	854
N2	20533	20595
N	21047	21210

Table 1: Number of participants across vaccine and placebo groups. N is the total number of participants in the specified group, N1 is the number of confirmed COVID-19 cases and N2 is the number of participants monitored during the entire surveillance period. *Adapted from Table 15 of the official FDA memorandum [1].*

Posterior distribution of the VE

We firstly infer the posterior distribution of VE assuming independent binomial risks for both, vaccine and placebo groups, as given in equations (1) and (2). Using the values from Table 1 we have that

$$\begin{aligned} Y_V = 81 &\sim \text{Binomial}(n_V = 20533, p_V), \\ Y_P = 854 &\sim \text{Binomial}(n_P = 20595, p_P). \end{aligned} \tag{6}$$

For a sequence of independent Bernoulli trials with some parameter $p \in [0, 1]$ that represents the probability of success (in our case the risk of infection), the likelihood function to consider is given by

$$P(k \mid p, n) = \text{Binomial}(n, p) = \binom{n}{k} p^k (1-p)^{n-k}. \tag{7}$$

Since the Beta distribution is conjugate to the Bernoulli likelihood, we use a Beta prior $Beta(\alpha, \beta)$, for which the posterior is given by

$$\begin{aligned} P(p \mid k, n) &\propto p^{k(1-p)^{n-k} p^{\alpha-1} (1-p)^{\beta-1}} \\ &\propto p^{k+\alpha-1}(1-p)^{\beta+n-k-1} \\ &\propto Beta(k+\alpha, n-k+\beta) \end{aligned} \tag{8}$$

In this work we set both α and β equal to 1, since this corresponds to the most common non-informative choice, also used for simplicity. From the above, we can infer that

$$\begin{aligned} p_V &\sim \text{Beta}(Y_V + 1, n_V - Y_V + 1) = \text{Beta}(20534, 20453), \\ p_P &\sim \text{Beta}(Y_P + 1, n_P - Y_P + 1) = \text{Beta}(20596, 19742), \end{aligned} \tag{9}$$

and the VE is given by

$$\text{VE} \sim 1 - \frac{\text{Beta}(20534, 20453)}{\text{Beta}(20596, 19742)}. \tag{10}$$

Since equation (10) does not follow a simple distribution, we approximate the posterior of VE by a Monte Carlo simulation. We draw a large number S of samples from the posterior distributions of p_V and p_P. This is done using *rbeta* in R, which generates the S samples following the distribution. Then, we compute the corresponding VE for each pair of samples, as implemented in the following code.

Listing 1: R script for the posterior distribution of VE

```
# Parameters
n_V <- 20533
y_V <- 81
n_P <- 20595
y_P <- 854


# set the number of samples on 100000 as an arbitrary number
set.seed(123)
S <- 100000

# rbeta to generate the S samples following the distribution
```

```
p_V <- rbeta(S,y_V+1,n_V-y_V+1)
p_P <- rbeta(S,y_P+1,n_P-y_P+1)

# computation of the VE for each pair of samples
VE <- 1-(p_V/p_P)
```

Figure 1 shows the posterior distribution of VE for the first model. The distribution is centered around 0.9037, which is the mean value as represented by the continuous black line in the plot. Moreover, the two dotted lines indicate the centered 95% credible interval, which is $[0.8807, 0.9241]$.

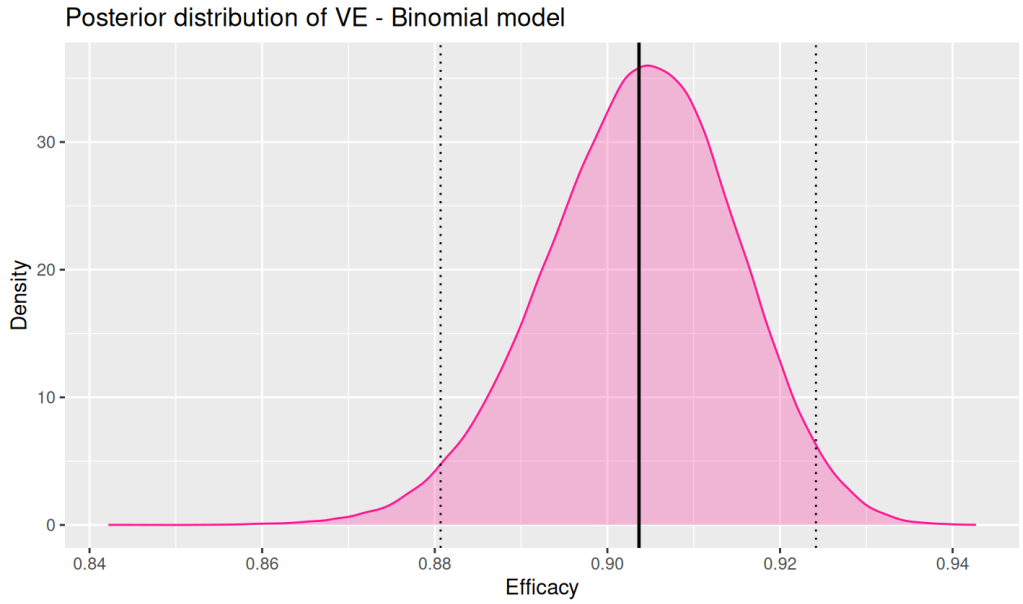


Figure 1: Posterior distribution of the Vaccine Efficacy for the Model 1 (Binomial). The continuous black line indicates the mean, while the two dotted lines indicate the centered 95% credible interval.

Model comparison

The second model was implemented in *Stan* via the *cmdstanr* interface for R. The model code is the following.

Listing 2: R script for the posterior distribution of VE using the logistic model

```
# Parameters
n_V<- 20533
y_V<- 81
n_P<- 20595
y_P<- 854

library(cmdstanr)

# Stan model
stan_code1<-"
data {
  int<lower=1> n_V;
  int<lower=1> n_P;
  int<lower=0> y_V;
  int<lower=0> y_P;
}
parameters {
  real alpha;
```

```
  real beta;
}
model {
  alpha ~ normal(0,5);
  beta  ~ normal(0,5);

  y_P ~ binomial_logit(n_P, alpha);
  y_V ~ binomial_logit(n_V, alpha + beta);
}
generated quantities {
  real p_V = inv_logit(alpha + beta);
  real p_P = inv_logit(alpha);
  real VE  = 1 - p_V / p_P;
}
"
writeLines(stan_code1, "binomial_logit.stan")
mod<- cmdstan_model("binomial_logit.stan")

data_list <- list(n_V = n_V, n_P = n_P, y_V = y_V, y_P = y_P)

fit <- mod$sample(
  data = data_list,
  seed = 123,
  chains = 4,
  parallel_chains = 4,
  iter_warmup = 1000,
  iter_sampling = 4000
)

draws <- fit$draws(format = "df")

VE_summary <- fit$summary("VE", "mean", ~quantile(.x, probs = c(0.025, 0.975)))
print(VE_summary)
```

The data block declares the sample sizes for the two arms, n_V and n_P, and the observed number of cases, y_V and y_P. The parameters block declares the two regression coefficients α and β. The model specifies Gaussian priors for both α and β and the likelihood via the *binomial_logit* function, applied separately to the placebo group $y_P \sim binomial_logit(n_P, \alpha)$ and the vaccine group $y_V \sim binomial_logit(n_V, \alpha + \beta)$ as suggested in equations (3) and (4). At the end, the function `writeLines` saves the model to a `.stan` file, which is then compiled with *cmdstan_model*. Then, four chains were run in parallel, each with 1000 warm-up iterations and 4000 sampling iterations, for a total of 16000 posterior draws.

Figure 2 shows the posterior distribution of VE for the second model. The distribution is centered around 0.9046 and the 95% credible interval is $[0.8820, 0, 9246]$. Table 2 summarize the comparison between the two models. We can conclude that both models yield to nearly identical posterior estimates, with means of 0.9037 for the Binomial model and 0.9046 for the Logistic model, differing only on the third and fourth significant digits. Also, the corresponding credible intervals overlap almost entirely, so the choice of model has an almost negligible impact on the resulting VE distribution.

This is expected as both models have independent Binomial likelihoods for the vaccine and placebo arms. This is particularly relevant, since the large sample size makes the likelihood highly informative relative to the priors (recall the influence of the prior is most pronounced in the low data regime [3]). This is a clear manifestation of the Bernstein-von Mises theorem, with sufficient data, Bayesian inference becomes prior-independent [3].

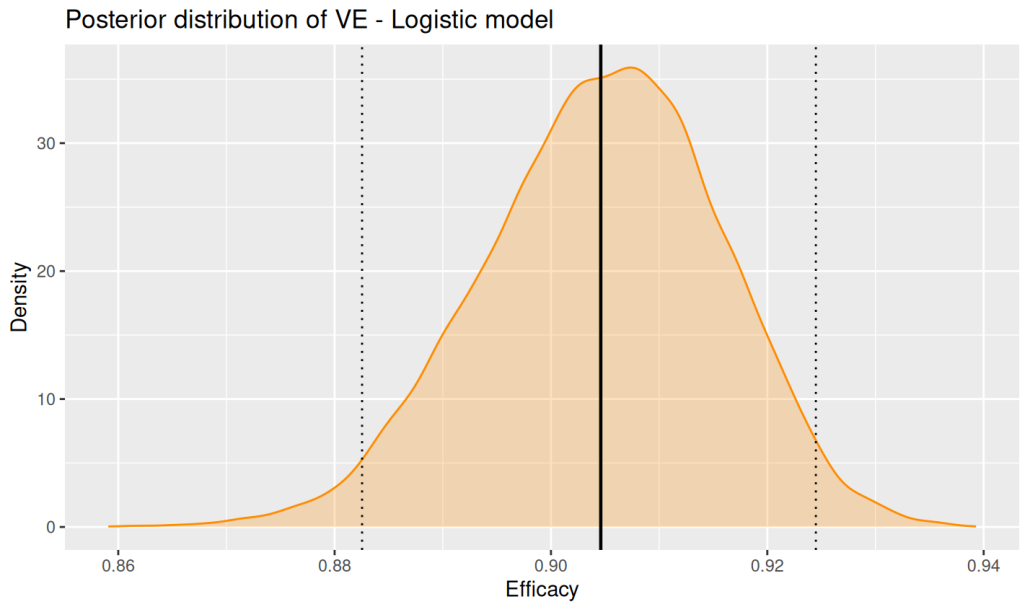


Figure 2: Posterior distribution of the Vaccine Efficacy for the Model 2 (Logistic).

	Model 1	**Model 2**
Model	Binomial	Logistic
Mean	0.9037	0.9046
Credible interval	[0.8807, 0.9241]	[0.8820, 0.9246]

Table 2: Summary of model comparison.

Posterior predictive distribution in a hypothetical trial

Finally, as both parameters p_P, p_V have been estimated, the posterior predictive distribution can be computed for a hypothetical trial with 10^4 participants per arm. This is done under the estimation of Model 1. Therefore, for each posterior draw p_P, p_V already obtained via `rbeta` in R, the number of cases in each arm is obtained as

$$\begin{aligned} Y_{V\,pred} &\sim \text{Binomial}(N = 10^4,\, p_V), \\ Y_{P\,pred} &\sim \text{Binomial}(N = 10^4,\, p_P) \end{aligned} \tag{11}$$

The implemented code is the following and the expected number of cases in each arm is of approximately 40 in the vaccinated and 415 in the placebo group.

Listing 3: R script for the posterior predictive distribution of cases in a hypothetical trial

```
# As we have determinated the parameter p_i(risk of disease),
# we can compute the posterior predictive distribution of cases
# in a hypothetical trial with 20000 participants (10000 per arm)

N=10000
p_V<- rbeta(S,y_V+1,n_V-y_V+1)
p_P<- rbeta(S,y_P+1,n_P-y_P+1)

y_V_pred<- rbinom(S, size = N, prob = p_V)
y_P_pred<- rbinom(S, size = N, prob = p_P)

```

```
cat("Expected cases with vaccine:", mean(y_V_pred)) #39.92
cat("Expected cases with placebo:", mean(y_P_pred)) #415.04
```

References

[1] Wollersheim, S., & Schwartz, A. (2021). *Clinical review memorandum: BLA 125742/0, COMIRNATY (COVID-19 vaccine, mRNA)*. Available at: `https://www.fda.gov/media/152256/download`

[2] Chu, H., & Halloran, M. E. (2004). Bayesian estimation of vaccine efficacy. Clinical Trials, 1(3), 306-314.

[3] Falasco, G. (2026). Advanced statistics for physics analysis: Bayesian inference [lecture notes].

Full R code

Listing 4: R script for the full exercise

```r
# Definition of the known parameters for our distribution calculation

# ****************************
#          BINOMIAL
# ****************************


# n_V: Number of persons with vaccine
# y_V: COVID 19 positive cases that were vaccinated
# n_P: Number of persons with placebo
# y_P: COVID 19 positive cases with placebo

n_V<- 20533
y_V<- 81
n_P<- 20595
y_P<- 854

# As we want to determinate the distribution of VE we have to
# calculate S random numbers over a beta distribution with the parameters n and y
# thats the reason why we will compare distributions!!!

# We define the number of samples on 100.000 as an arbitrary number
#
set.seed(42)
S<- 100000
# As the likelihood is a binomial distribution, we are going to implement a
# prior Beta(a,b), in that order, for an uniform and not informed
# Beta(1,1) we have it in the form the posterior defined as:
# Beta(alpha + y, beta+ n  - y):

p_V<- rbeta(S,y_V+1,n_V-y_V+1)
p_P<- rbeta(S,y_P+1,n_P-y_P+1)

#Lets remember that rbeta (r function) does the empirical normalization
# Now as defined the VE , we will compute it and have the distributed values of it

VE<- 1-(p_V/p_P)

# In this way, the mean value of the distribution of the Vaccine Eficciency will be :
mVE<- mean(VE)
cat('Mean VE :', mVE)

# For the 95% of credible interval we implement the quantile calculation over the distribution:
q95<- quantile(VE,c(0.025,0.975))
cat('Credible interval: ',q95)



library(ggplot2)


df <- data.frame(VE)

#
ggplot(df, aes(VE)) +
  geom_density(color = 'deeppink', fill = 'deeppink',alpha = 0.25) +
  geom_vline(xintercept = mean(VE), color = "black", size = 0.8) +
  geom_vline(xintercept = quantile(VE, c(0.025, 0.975)), color = "black", linetype = "dotted")+
  labs(title = "Posterior distribution of Vaccine Eficcacy (VE)",
    x = " Eficacy",
    y = "Density"
  )

# ***************************
#  Logistic
# ***************************

# Implementing the Logistic regression in Stan , we have then :

library(cmdstanr)
stan_code1<-"
data {
  int<lower=1> n_V;
  int<lower=1> n_P;
  int<lower=0> y_V;
  int<lower=0> y_P;
}
parameters {
  real alpha;
  real beta;
}
model {
  alpha ~ normal(0,5);
  beta  ~ normal(0,5);

  y_P ~ binomial_logit(n_P, alpha);
  y_V ~ binomial_logit(n_V, alpha + beta);
}
generated quantities {
  real p_V = inv_logit(alpha + beta);
  real p_P = inv_logit(alpha);
  real VE  = 1 - p_V / p_P;
}
```

```
"
writeLines(stan_code1, "binomial_logit.stan")
mod<- cmdstan_model("binomial_logit.stan")

data_list <- list(n_V = n_V, n_P = n_P, y_V = y_V, y_P = y_P)

fit <- mod$sample(
  data = data_list,
  seed = 123,
  chains = 4,
  parallel_chains = 4,
  iter_warmup = 1000,
  iter_sampling = 4000
)

draws <- fit$draws(format = "df")

VE_summary <- fit$summary("VE", "mean", ~quantile(.x, probs = c(0.025, 0.975)))

print(VE_summary)

df <- data.frame(VE = draws$VE)

ggplot(df, aes(VE)) +
  geom_density(color = 'darkorange', fill = 'darkorange', alpha = 0.25) +
  geom_vline(xintercept = mean(draws$VE), color = "black", linewidth = 0.8) +
  geom_vline(xintercept = quantile(draws$VE, c(0.025, 0.975)), color = "black", linetype = "dotted") +
  labs(title = "Posterior distribution of VE - Logistic model",
       x = "Efficacy",
       y = "Density")

cat(mean(draws$VE), quantile(draws$VE,c(0.025,0.975)))


# ********************
# New estimation
# ********************

# As we have determinate the parameter p_i(risk of disease),
#we can compute the posterior predictive
# distribution of cases in a hypothetical trial with 20000 participants (10000 per arm)
N=10000

p_V<- rbeta(S,y_V+1,n_V-y_V+1)
p_P<- rbeta(S,y_P+1,n_P-y_P+1)

# As in the first point, then,

y_V_pred<- rbinom(S, size = N, prob = p_V)
y_P_pred<- rbinom(S, size = N, prob = p_P)



cat("Expected cases with vaccine:", mean(y_V_pred))
cat("Expected cases with placebo:", mean(y_P_pred))
```